arXiv:2601.16965v1 [cs.AI] 23 Jan 2026

A promising direction in GIScience suggests that geo-analytical questions encode implicit procedu-

sive capabilities in tool-augmented reasoning. ReAct (Yao et al., 2022) introduced the thought-action-observation loop for interleaved reasoning and acting. Toolformer (Schick et al., 2023) enables self-supervised tool use learning, while CodeAct (Wang et al., 2024) shows that generating executable code outperforms JSON-based actions by up to 20%. In the geospatial domain, LLM-Geo (Li and Ning, 2023) and GeoGPT (Zhang et al., 2024)



3.4 Retrieval-augmented Orchestration

Method	Overall	Place Info	Nearby	Routing	Trip
Closed-Source LLMs					
Direct LLM (GPT-4o-mini)	32.23	45.31	32.53	22.73	28.36

periments demonstrate that Spatial-Agent significantly outperforms existing agent baselines. Error analysis reveals that the primary bottleneck lies in external API interactions rather than reasoning components, validating the effectiveness of

Zekun Li, Malcolm Grossman, Mihir Kulkarni, Muhao Chen, Yao-Yi Chiang, and 1 others. 2025. Mapqa: Open-domain geospatial question answering on map data. *arXiv preprint arXiv:2503.07871*.

Zhenlong Li and Huan Ning. 2023. Autonomous gis: the next-generation ai-powered gis. *International Journal of Digital Earth*, 16(2):4668–4686.

Gengchen Mai, Krzysztof Janowicz, Ling Cai, Rui Zhu, Blake Regalia, Bo Yan, Meilin Shi, and Ni Lao. 2020. Se-kge: A location-aware knowledge graph embedding model for geographic question answering and spatial semantic lifting. *Transactions in GIS*

B.3 Procedural Precedence

Procedural roles follow a strict execution order defined as:

$$\text{SUBCOND} \quad \text{COND} \quad \text{SUPPORT} \quad \text{MEASURE}: \tag{7}$$

This ordering reflects the semantics of geoanalytical workflows: constraints are applied before supports are constructed, and measurements are computed last.

B.4 Examples

Table 5

where $a = \sin^2 \frac{}{2} + \cos(\ _1)\cos(\ _2)\sin^2 \frac{}{2}$ and $R = 6371$ km is Earth's mean radius.

bearing($_1$**,** λ_1**,** $_2$**,** λ_2**)** Computes the initial bearing (forward azimuth) from point 1 to point 2, returned as degrees clockwise from true north (0 –360).

bearing_to_direction(θ**)** Converts a numeric bearing θ